



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

01-01-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS, TAAS, TEAS, TIAS, TMAS, I(A), LIVE-I(A) & III] (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	A	A	C	B	C	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	C	C	C	A	C	B	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4	600	8	140	2	1	9	2	5	6

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	C	A	D	D	D	D	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	D	B	C	D	D	A	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4	1	3	56	300	85	32	4	9	5

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	A	C	C	D	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	D	D	C	C	D	C	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	5	3	2	5	6	5	5	7	0

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

In new system,

$$\text{Force unit} = \frac{(10\text{kg})(5\text{m})}{(1\text{sec})^2} = 50 \text{ kgm/s}^2 = 50 \text{ N}$$

2. Ans (A)

$$10 \times 4 + \frac{1}{2} \times 4 \times 10 + \frac{1}{2} \times 2 \times 5$$

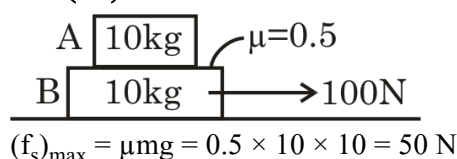
$$= 40 + 20 + 5 = 65 \text{ m}$$

3. Ans (C)

$$V_{\text{net}} = \sqrt{4^2 + 4^2 + 2 \times 4 \times 4 \cos 120^\circ}$$

$$= 4 \text{ m/s}$$

6. Ans (C)



Assuming they move together,

$$100 = 20a \Rightarrow a = 5 \text{ m/s}^2$$

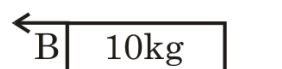
Check friction on A :

$$f = 10 \times 5 = 50 \text{ N}$$

$$f_{\text{regd}} = f_{\text{available}}$$

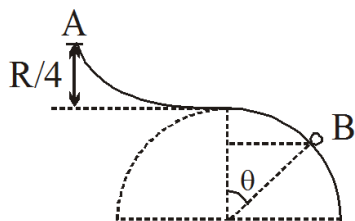
 $\Rightarrow$ They move together

$$f_r = 50 \text{ N}$$



$$W_{fe} = -50 \times 2 = -100 \text{ J}$$

8. Ans (C)



Conserving energy between points A & B,

$$mg \left[\frac{R}{4} + R(1 - \cos \theta) \right] = \frac{1}{2} mv^2$$

Also EFD at point B will give,

$$mg \cos \theta = \frac{mv^2}{R}$$

$$\therefore \frac{1}{4} + (1 - \cos \theta) = \frac{1}{2} \cos \theta$$

$$\frac{5}{4} = \frac{3}{2} \cos \theta$$

$$\therefore \cos \theta = \frac{5}{6}$$

10. Ans (C)

$$P_i = mv_0 \hat{j}$$

After shoots momentums of system will remain conserved.

$$\vec{P}_f = \frac{2}{3} m\vec{v} + \frac{m}{3} (2v_0) \hat{i}$$

$$\vec{P}_i = \vec{P}_f$$

$$mv_0 \hat{j} = \frac{2}{3} m\vec{v} + \frac{2}{3} mv_0 \hat{i}$$

$$\vec{v} = \frac{3}{2} v_0 \hat{j} - v_0 \hat{i}$$

$$|\vec{v}| = \sqrt{\frac{9}{4} v_0^2 + v_0^2} = \frac{\sqrt{13}}{2} v_0$$

20. Ans (A)

Avoiding end correction, the length of closed organ pipe is

$$\ell_2 = \frac{\lambda_1}{4} \text{ or } \lambda_1 = 4\ell_1$$

The length of open organ pipe is

$$\ell_2 = \frac{\lambda_2}{2} \text{ or } \lambda_2 = 2\ell_2$$

Here $n_1 = n_2$

$$\Rightarrow \frac{v}{\lambda_1} = \frac{v}{\lambda_2} \text{ or } \frac{v}{4\ell_1} = \frac{v}{4\ell_2}$$

Therefore $\ell_1 : \ell_2 = 1 : 2$

PART-1 : PHYSICS

SECTION-II

1. Ans (4)

$$a_{\text{com}} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$$

$$\frac{20}{11} = \frac{8a - 3 \times 4}{11}$$

$$\Rightarrow a = \frac{4}{m} \frac{m}{s^2}$$

2. Ans (600)

No external force in horizontal direction

Hence, momentum conservation

$$5 \times 10^3 \times 1.2 = (5+1) \times 10^3 \times v$$

$$v = 1 \text{ m/s}$$

$$\therefore K_i = \frac{1}{2} \times 5 \times 10^3 \times (1.2)^2$$

$$\therefore K_f = \frac{1}{2} \times 6 \times 10^3 \times (1)^2$$

$$\Delta K = 600 \text{ J}$$

3. Ans (8)

Torque balance

$$m_1 \ell_1 = m_2 \ell_2$$

$$16\ell_1 = m\ell_2$$

$$m\ell_1 = 4\ell_2$$

$$m^2 = 64 \quad m = 8 \text{ kg.}$$

4. Ans (140)

$$\text{Given that } \frac{2}{5} MR^2 = 40 \text{ or } MR^2 = 100$$

by theorem of parallel axes

$$I = I_{\text{CM}} + MR^2 = \frac{2}{5} MR^2 + MR^2 = \frac{7}{5} MR^2$$

$$= \frac{7}{5} \times 100 = 140 \text{ kg-m}^2$$

5. Ans (2)

$$1 \times S_A (30 - 25) = 2 \times S_B \times (25 - 20)$$

$$\frac{S_A}{S_B} = 2$$

7. **Ans (9)**

In steady state situation no current will flow through the capacitor, 2Ω and 3Ω are in parallel.

Therefore, their combined resistance will be

$$R = \frac{2 \times 3}{2 + 3} = 1.2 \Omega$$

$$\text{Net current through the battery: } i = \frac{6}{1.2 + 2.8} = 1.5 \text{ A}$$

This current will distribute in inverse ratio of their resistance 2Ω and 3Ω .

$$\therefore \frac{i_2}{i_3} = \frac{3}{2}$$

$$\text{or } i_2 = \left(\frac{3}{3+2} \right) (1.5) = 0.9 \text{ A}$$

10. **Ans (6)**

$$F = \eta A \frac{v}{d}$$

$$\frac{F_1}{F_2} = \frac{v_1}{v_2}$$

$$\frac{800}{2400} = \frac{2}{v_2}$$

$$v_2 = 6 \text{ cm/s}$$

PART-2 : CHEMISTRY

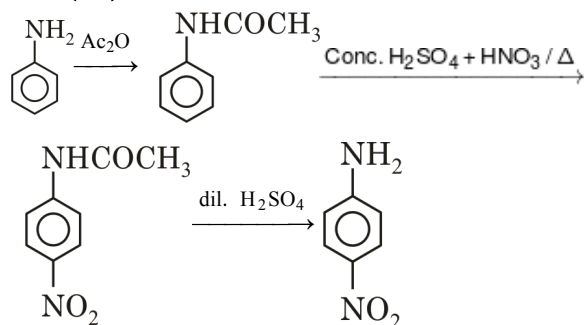
SECTION-I

1. **Ans (A)**

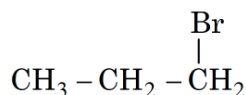
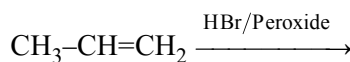
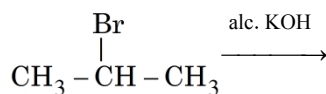
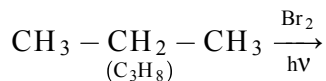
If R is $-\text{CH}_3$ (lowest alkyl) then above method gives amino acid of more than 2-carbon atoms while glycine is



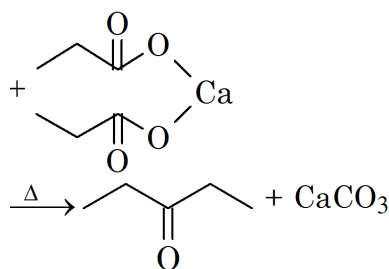
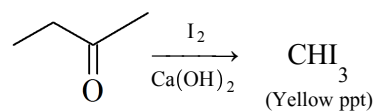
2. **Ans (B)**



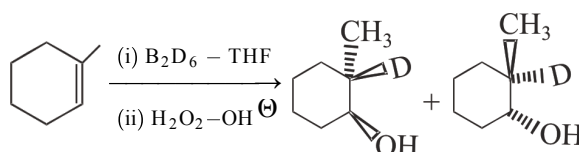
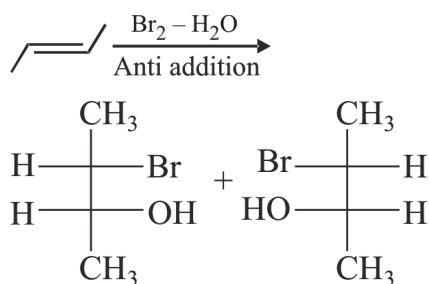
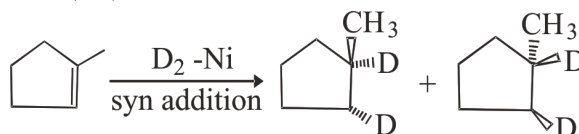
3. **Ans (C)**



4. **Ans (A)**



5. **Ans (D)**



6. **Ans (D)**

MnO_2, Δ is used to oxidise allylic primary alcohol to aldehyde

8. **Ans (D)**

Let 100 gm solution

$$M = \frac{\frac{1}{120}}{\frac{100}{1-2}} \times 1000 = 0.1 \text{ M NaA}$$

$$\begin{aligned} \text{As pH} &= 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log C \\ &= 7 + \frac{1}{2} + \frac{1}{2} \log 0.1 = 9.5 \end{aligned}$$

9. **Ans (D)**

$$50 = \frac{4 \times 60}{6 \times 10^{23} \times (a \times 10^{-8})^3}$$

$$a = 2\text{\AA}$$

$$\text{Distance between two Na}^+ \text{ ions} = \frac{a}{\sqrt{2}} = \sqrt{2}\text{\AA}$$

10. **Ans (C)**

$$\text{For Glucose (i}_1\text{C}_1) = 1 \times 0.3 = 0.3$$

$$\text{For NaCl (i}_2\text{C}_2) = 2 \times 0.2 = 0.4$$

$$\text{For Na}_3\text{PO}_4 \text{ (i}_3\text{C}_3) = 4 \times 0.3 = 1.2$$

$$\text{For Benzoic acid (i}_4\text{C}_4) =$$

$$\left[1 + 0.5 \left(\frac{1}{2} - 1 \right) \right] \times 0.4 = 0.3$$

11. **Ans (B)**

$$(A) = -76 - 2.37$$

$$(B) 0.76 + 0.8 = 1.56$$

$$(C) -0.34 + 0.8$$

$$(D) -0.8 - 2.37$$

$$= 1.57$$

12. **Ans (D)**

$$\frac{E_2(\text{HCl})}{E_3(\text{Li})} = \frac{-13.6 \times \frac{2^2}{2^2}}{-13.6 \times \frac{3^2}{3^2}} = 1$$

$$\frac{r_1(\text{x})}{r_2(\text{Be}^{+3})} = \frac{0.529 \times \frac{12}{1}}{0.54 \times \frac{3^2}{3}} = 1$$

13. **Ans (D)**

As volume increases, reaction moves on that side where gaseous moles are more.

$$\text{As volume increases, [concentration]} = \frac{\text{moles}}{\text{volume}}$$

14. **Ans (B)**

[Spin magnetic moment $\propto$ No. of unpaired electrons]

$$(1) [\text{Cr}(\text{CN})_6]^{3-} \Rightarrow (t_{2g})^3 (e_g)^0 = 3 \text{ u.p.e}$$

$$(2) [\text{FeF}_6]^{3-} \Rightarrow (t_{2g})^3 (e_g)^2 = 5 \text{ u.p.e.}$$

$$(3) [\text{Ni}(\text{en})_3]^{2+} \Rightarrow (t_{2g})^6 (e_g)^2 = 2 \text{ u.p.e}$$

$$(4) [\text{Cu}(\text{NH}_3)_6]^{2+} \Rightarrow (t_{2g})^6 (e_g)^3 = 1 \text{ u.p.e}$$

PART-2 : CHEMISTRY

SECTION-II

4. **Ans (56)**

$$\Delta G^0 = \Delta H_0 - T\Delta S^0$$

$$\Delta G^0 = 75000 - 300 \times 300 = -15000 \text{ J/mole}$$

$$\Delta G = \Delta G^0 + RT \ln Q$$

$$\Delta G = -15000 + 8 \times 300 \ln$$

$$\frac{\left(\frac{1}{2}\right)^2}{\frac{1}{2}} = -16680 \text{ J/mol}$$

$$-T\Delta S_{\text{universe}} = \Delta G$$

$$\Delta S_{\text{universe}} = \frac{-16680}{-300} = 55.60$$

5. **Ans (300)**

$$A \rightarrow B$$

$$K_1 = A_1 \times e^{\frac{-E_{a1}}{RT}}$$

$$x \rightarrow y$$

$$K_2 = A_2 \times e^{\frac{-E_{a2}}{RT}}$$

$$\text{if } k_1 = k_2$$

$$10^9 e^{\frac{-1200}{RT}} = 10^{10} e^{\frac{-1800}{RT}}$$

$$\frac{-1200}{RT} = \frac{-1800}{RT} + \ln 10$$

$$T = \frac{300}{2.303} \text{ K}$$

6. **Ans (85)**

$$P_1 = 5 \text{ cm of Hg}$$

$$V_1 = \pi r^2 (10)$$

$$P_2 = ?$$

$$V_2 = \pi r^2 (5)$$

$$\text{So, } P_2 = 10 \text{ cm of Hg}$$

$$\text{Atmospheric pressure} = 85 \text{ cmHg.}$$

7. Ans (32)

$$P_C = \frac{a}{27b^2} = 100$$

$$T_C = \frac{8a}{27Rb} = 300$$

$$b = \frac{3R}{8} = \frac{1}{32}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

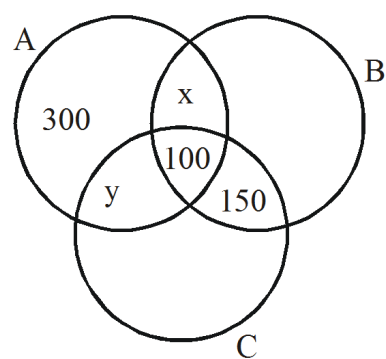
$$\text{Let } x_i - 5 = d_i$$

$$\begin{aligned}\sigma_x^2 &= \sigma_d^2 = \frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n} \right)^2 \\ &= \frac{125}{10} - \left(\frac{5}{10} \right)^2 = \frac{25}{2} - \frac{1}{4} = \frac{49}{4}\end{aligned}$$

2. Ans (B)

$$S \equiv (\sim p \wedge r) \vee (\sim p \vee p) \equiv (\sim p \wedge r) \vee t \equiv (\sim p \wedge r)$$

3. Ans (B)

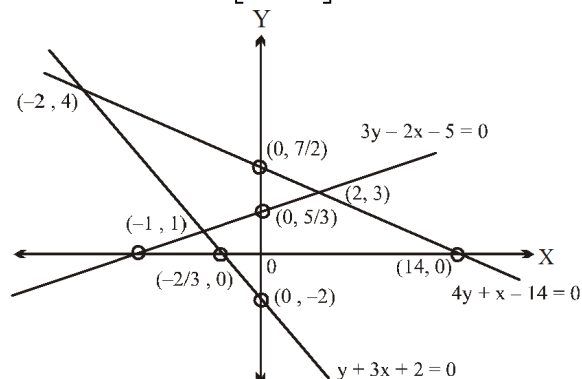


number of people who read exactly two newspapers

$$= x + y + 150 = 100 + 150 = 250$$

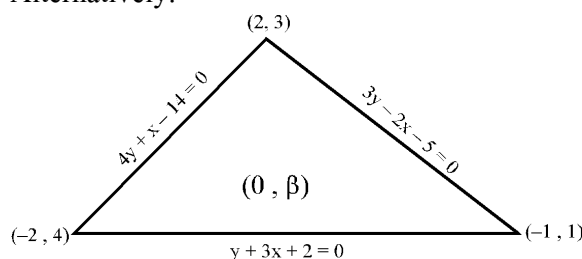
4. Ans (A)

$$\{2, 3\}, \text{ where } k \in \left[\frac{5}{3}, \frac{7}{2} \right]$$



Obviously $5/3 \leq k \leq 7/2$

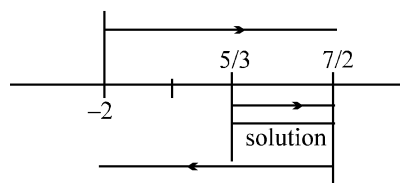
Alternatively:



Substituting (2, 3) in $y + 3x + 2 = 0$

$$\text{we get } 13 > 0 \Rightarrow b + 2 \geq 0$$

$$\text{similarly } 3\beta - 5 \geq 0 \Rightarrow 5/3 \geq \beta \geq 7/2$$



$$\text{and } (2\beta - 7) \leq 0$$

5. Ans (C)

$$\begin{aligned}A &= \int_0^{\pi} \frac{\sin x}{x^2} dx \\ &= \left(\sin x \left(-\frac{1}{x} \right) \right)_0^{\pi} + \int_0^{\pi} \frac{\cos x}{x} dx \\ A &= 0 - (-1) + \int_0^{\pi} \frac{\cos x}{x} dx\end{aligned}$$

Put $x = 2y$

$$A = 1 + \int_0^{\pi/2} \frac{\cos 2x}{x} dx$$

$$\int_0^{\pi/2} \frac{\cos 2x}{x} dx = A - 1$$

6. **Ans (C)**

$$\begin{vmatrix} \tan \theta & \cot \theta & 8 \cos 2\theta \\ \cot \theta & 8 \cos 2\theta & \tan \theta \\ 8 \cos 2\theta & \tan \theta & \cot \theta \end{vmatrix} = 0$$

$$\Rightarrow \tan \theta = \cot \theta = 8 \cos 2\theta \text{ (not possible)}$$

$$\text{or } \tan \theta + \cot \theta + 8 \cos 2\theta = 0$$

$$\Rightarrow \frac{1}{\sin \theta \cos \theta} = -8 \cos 2\theta \Rightarrow \sin 4\theta = -\frac{1}{2}$$

7. **Ans (D)**

curves are confocal

$\therefore$ orthogonal.

8. **Ans (B)**

$$\lim_{n \rightarrow \infty} \prod \left(1 + \frac{2}{k^2 + 5k + 4} \right)$$

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \frac{k^2 + 5k + 6}{k^2 + 5k + 4}$$

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \frac{(k+2)(k+3)}{(k+1)(k+4)}$$

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(\frac{k+2}{k+1} \right) \prod_{k=1}^n \left(\frac{k+3}{k+4} \right)$$

$$\lim_{n \rightarrow \infty} \left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \dots \times \frac{n+2}{n+1} \right) \left(\frac{4}{5} \times \frac{5}{6} \times \frac{6}{7} \times \dots \times \frac{n+3}{n+4} \right)$$

$$= 2$$

9. **Ans (B)**

$$|A \cdot \text{Adj}(A)| = |A| \cdot |\text{Adj}(A)| = |A| \cdot |A|^2 = |A|^3$$

$$\text{here } |A| = abc - 2(10a + 3b + c) + 34$$

$$= -10 - 2(17) + 34 = -10$$

$$\therefore |A|^3 = -1000$$

10. **Ans (A)**

$$|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$$

$$|\text{adj}(\text{adj}(\text{adj}(A)))| = |A|^{(n-1)^3} = 3^8$$

11. **Ans (C)**

$$f(x) = \frac{(\sin \pi x)(x^2 + 9)(e^{x-2} + 5)}{(5 + x^3)(1 + x^2)(2 + x)}$$

$$\text{Let } f(x) = (\sin \pi x)(g(x))$$

$$= f'(x) = (\sin \pi x) \cdot g'(x) + g(x) \cdot \pi \cdot (\cos \pi x)$$

$$g(2) = \frac{(13)(6)}{(13)(5)(4)} = \frac{3}{10}$$

$$\Rightarrow f'(2) = 0 + \frac{3\pi}{10} = \frac{3\pi}{10}$$

12. **Ans (B)**

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{\frac{h-h \cdot 3^{\frac{1}{h}}}{3^{\frac{1}{h}}-1}}{h} = 0$$

$$= \lim_{h \rightarrow 0} \frac{h \cdot 3^{\frac{1}{h}}}{h \left(3^{\frac{1}{h}} - 1 \right)}$$

$$= \lim_{h \rightarrow 0} \frac{3^{\frac{1}{h}}}{3^{\frac{1}{h}} - 1} = 1$$

$$\& f'(0^-) = \lim_{h \rightarrow 0} \frac{\frac{-h+h \left(3^{-\frac{1}{h}} + 1 \right)}{3^{-\frac{1}{h}} - 1}}{-h} = 0$$

$$= \lim_{h \rightarrow 0} \frac{h \cdot 3^{-\frac{1}{h}}}{h \left(1 - 3^{-\frac{1}{h}} \right)}$$

$$= \lim_{h \rightarrow 0} \frac{3^{-\frac{1}{h}}}{1 - 3^{-\frac{1}{h}}} = 0$$

13. **Ans (D)**

$$\text{sgn}(x^2 - x + 1) = \text{sgn} \left(\left(x - \frac{1}{2} \right)^2 + \frac{3}{4} \right) = 1$$

$$\text{for } x \in [0, 3], x^2 - x + 1 \in \left[\frac{3}{4}, 7 \right]$$

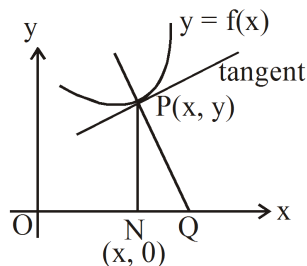
$$[x^2 - x + 1] \text{ is discontinuous where } x^2 - x + 1$$

$$= 1, 2, 3, 4, 5, 6, 7$$

$$\text{but } x^2 - x + 1 = 1 \Rightarrow x = 0, 1$$

$\therefore$ Discontinuous at 8 points.

14. Ans (D)



Equation of normal at P(x, y) is

$$Y - y = -\frac{dx}{dy}(X - x)$$

It meets the x-axis when $X = y \frac{dy}{dx} + x$

$$\Rightarrow OQ = x + y \frac{dy}{dx}$$

$$\text{Also, } ON = x \Rightarrow NQ = y \frac{dy}{dx}$$

$$\Rightarrow x \frac{(1+y^2)}{1+x^2} = y \frac{dy}{dx} \Rightarrow \frac{xdx}{1+x^2} = \frac{ydy}{1+y^2}$$

$$\Rightarrow \int \frac{2xdx}{1+x^2} = \int \frac{2ydy}{1+y^2}$$

$$\text{or } \log(1+x^2) = \log(1+y^2) + C \Rightarrow \frac{1+x^2}{1+y^2} = k$$

As the curve passes through (3, 1) therefore

$$k = 5$$

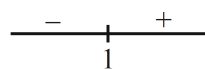
$$\text{Hence the equation is } x^2 - 5y^2 = 4$$

15. Ans (C)

$$\text{Let } f(x) = (x-1)^2 + x \ln x - x$$

Domain : $x > 0$

$$f'(x) = 2(x-1) + \ln x$$



$$\because f(1) = -1, f(0^+) = 1, f(\infty) \rightarrow \infty$$

$f(x) = 0$ has exactly two roots.

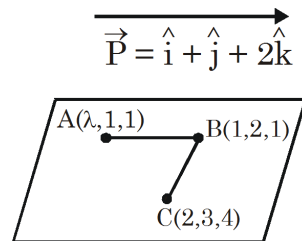
16. Ans (C)

$$\because \text{line lies in the plane} \Rightarrow 2 - 2\lambda + k = 0$$

$$\text{and } 2 - 3 + 2\lambda = 0$$

$$\Rightarrow \lambda = \frac{1}{2} \text{ and } k = -1.$$

17. Ans (D)



$$\vec{BA} = (\lambda - 1) \hat{i} - \hat{j}$$

$$\vec{BC} = \hat{i} + \hat{j} + 3\hat{k}$$

$$\left[\vec{BA} \quad \vec{BC} \quad \vec{P} \right] = 0$$

$$\Rightarrow \begin{vmatrix} \lambda - 1 & -1 & 0 \\ 1 & 1 & 3 \\ 1 & 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 1)(2 - 3) + 1(2 - 3) = 0$$

$$-(\lambda - 1) - 1 = 0$$

$$\Rightarrow \lambda = 0$$

18. Ans (C)

$$\hat{a} \times \hat{b} + \hat{a} = \vec{c} \quad \dots(i)$$

$$\hat{a} \times \hat{b} = \vec{c} - \hat{a}$$

$$\text{Dot with } \vec{c} \Rightarrow v = [\hat{a} \quad \hat{b} \quad \vec{c}] = \vec{c}^2 - \hat{a} \cdot \vec{c} \quad \dots(ii)$$

for (i) : dot with $\hat{b}$

$$\Rightarrow 0 + \hat{a} \cdot \hat{b} = \hat{b} \cdot \vec{c} = 0 \Rightarrow \hat{a} \cdot \hat{b} = 0$$

$$\Rightarrow \hat{a} \perp \hat{b} \quad \dots(iii)$$

for (i) : squaring

$$(\hat{a} \times \hat{b})^2 + 1 + 2(0) = \vec{c}^2 \quad (\text{by (iii)})$$

$$\Rightarrow \vec{c}^2 = 1 + 1 = 2 \quad \dots(iv)$$

$$\text{for (i) : dot with } \hat{a} : \hat{a}^2 = \vec{c} \cdot \hat{a} \quad \dots(v)$$

put (iv) & (v) in (ii) :

$$v = 2 - 1 = 1$$

19. Ans (B)

$$z^7 - 7 = (z - z_0)(z - z_1) \dots (z - z_6)$$

Putting (1) $z = -z_0$

$$-14 = -z_0^7 - 7 = -2z_0(z_0 + z_1)(z_0 + z_2) \dots (z_0 + z_6)$$

(2) $z = -z_1$

$$-14 = -z_1^7 - 7 = -2z_1(z_1 + z_0)(z_1 + z_2) \dots (z_1 + z_6)$$

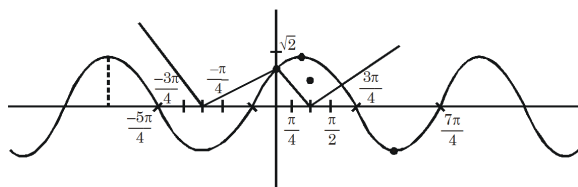
and similarly $z = -z_2, -z_3, \dots, -z_6$

Multiplying all such equations

$$-14^7 = -2^7(z_0 z_1 \dots z_6) \left(\prod_{0 \leq i < j \leq 6} (z_i + z_j) \right)^2$$

$$7^6 = \prod_{0 \leq i < j \leq 6} (z_i + z_j)^2 = p^2$$

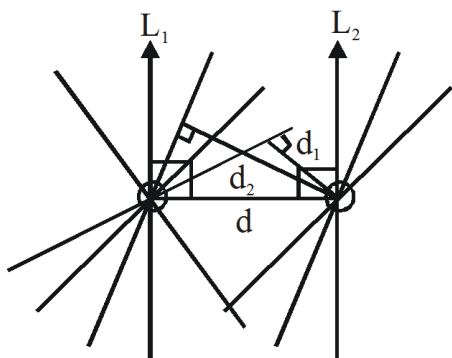
20. Ans (B)



PART-3 : MATHEMATICS

SECTION-II

1. Ans (5)



Maximum distance occurs if L_1 & L_2 are parallel to each other and perpendicular to line joining the points of concurrency. It is also obvious from figure as $d > d_2 > d_1 \dots$

2. Ans (5)

Point of parabola $y^2 = 2x$ is $\left(\frac{t^2}{2}, t\right)$

Normal at t is $y + tx - 2at - at^3 = 0$

$$y + tx - t - \frac{t^3}{2} = 0$$

But normal is given by

$$y - \sqrt{2}x - 14k\sqrt{2} = 0$$

$$\Rightarrow t = -\sqrt{2}$$

$$k = \frac{1}{2}$$

Point is $(1, -\sqrt{2})$

other end of normal chord is $(4, 2\sqrt{2})$

Length of normal chord $\sqrt{9+18} = \sqrt{27} = 3\sqrt{3}$

$$a = 3, k = \frac{1}{2}$$

3. Ans (3)

$$f(x) = \int e^x \left(\frac{2}{1 - \tan x} + \tan^2 \left(x + \frac{\pi}{4} \right) \right) dx$$

$$= \int e^x \left(\frac{1 + \tan x}{1 - \tan x} + \sec^2 \left(x + \frac{\pi}{4} \right) \right) dx$$

$$= \int e^x \left(\tan \left(\frac{\pi}{4} + x \right) + \sec^2 \left(x + \frac{\pi}{4} \right) \right) dx$$

$$\Rightarrow f(x) = e^x \tan \left(\frac{\pi}{4} + x \right) + C$$

$$\therefore f \left(\frac{3\pi}{4} \right) = 0 \Rightarrow C = 0$$

$$f(\pi) = e^\pi$$

$$\ln f(\pi) = \pi$$

$$[\ln f(\pi)] = 3$$

4. **Ans (2)**

$$\cos^{-1} \frac{p}{a} + \cos^{-1} \frac{q}{b} = \alpha$$

Hence $\cos \alpha$

$$= \cos \left(\cos^{-1} \frac{p}{a} \right) \cdot \cos \left(\cos^{-1} \frac{q}{b} \right)$$

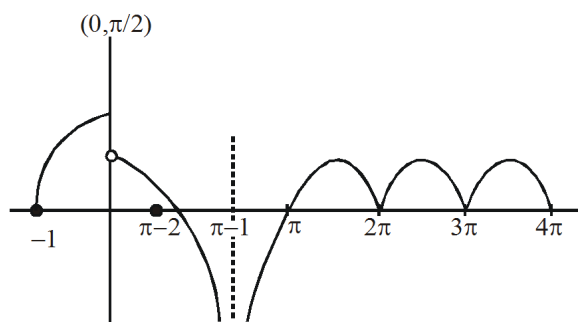
$$- \sin \left(\cos^{-1} \frac{p}{a} \right) \sin \left(\cos^{-1} \frac{q}{b} \right)$$

$$\cos \alpha = \frac{p}{a} - \frac{q}{b} - \sqrt{\frac{1-p^2}{a^2}} \sqrt{1 - \frac{q^2}{b^2}}$$

$$\left(\frac{pq}{ab} - \cos \alpha \right)^2 = \left(\frac{1-p^2}{a^2} \right) \left(1 - \frac{q^2}{b^2} \right)$$

$$\Rightarrow \frac{p^2}{a^2} - \frac{2pq}{ab} \cos \alpha + \frac{a^2}{b^2} = 1 - \cos^2 \alpha \in [0,1]$$

5. **Ans (5)**



$f(x)$ is non derivable at $x = -1, 0, \pi-1$ or $2\pi, 3\pi$

6. **Ans (6)**

$$\Rightarrow V = \pi r^3 + (2r)^3$$

$$\frac{dv}{dt} = (\pi + 8)3r^2 \cdot \frac{dr}{dt} = (\pi + 8) \cdot 3 \cdot \frac{4}{2(\pi + 8)}$$

$$= 6 \text{ cc/sec}$$

7. **Ans (5)**

$$\frac{{}^{20}C_1 \times {}^{13}C_5}{6}$$

8. **Ans (5)**

$$\text{Given } P(1) = K; P(2) = 2^2K; P(3) = 3^2K;$$

$$P(4) = 4^2K; P(5) = 5^2K; P(6) = 6^2K$$

$$\therefore \text{Total} = 91K = 1$$

$$\Rightarrow K = \frac{1}{91}; \therefore P(1) = \frac{1}{91}; P(2) = \frac{4}{91} \text{ and so on.}$$

Let three events A, B, C are defined as

$$A : a < b$$

$$B : a = b$$

$$C : a > b$$

By symmetry, $P(A) = P(C)$. Also $P(A) + P(B) +$

$$P(C) = 1 \quad \dots(1)$$

$$\text{Since } P(B) = \sum_{i=1}^6 [P(i)]^2 = \frac{25}{91}$$

$$2P(A) + P(B) = 1 \quad (\text{from (i)})$$

$$P(A) = \frac{1}{2} [1 - P(B)] = \frac{33}{91}$$

$$33 + 91 = 124.$$

9. **Ans (7)**

$$bc(a^2 + 4a + 4) = 81$$

$$\Rightarrow bc(a+2)^2 = 81$$

$$\frac{(a+2) + (a+2) + b + c}{4} \geq [(a+2)^2 bc]^{1/4}$$

$$2a + b + c + 4 \geq 12$$

Equality occurs when $a + 2 = b = c = 3$.

$$\Rightarrow a = 1, b = 3, c = 3$$

10. **Ans (0)**

$$(1 - e^x)^{100} = \sum_{r=0}^{100} (-1)^r {}^{100}C_r e^{rx} = f(x)$$

If $f^n(x)$ represents n^{th} derivative of $f(x)$; then

$$\text{required seires is } \frac{f^{50}(0)}{(50)^{100}} = 0.$$